

Arunabha Majumder

Mechanical R&D Engineer | Mechanisms & Wearable Robotics | Prototyping | Test Rigs & Validation

Aalborg, Denmark | +45 71 62 24 00 | somrkmv1997@gmail.com |
linkedin.com/in/arunabha-majumder-681264107

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Head of Innovation and R&D Lead
Levitate
Copenhagen, Denmark

Application: Mechanical R&D Engineer

Dear Hiring Team,

I am applying for the Mechanical R&D Engineer position. I hold a B.Tech in Mechanical Engineering and an M.Tech in Mechatronics, and I am finishing my PhD at Aalborg University, where I develop wearable mechanisms that carry load on the human body. Building devices that people wear and move with, and that have to be light, strong and predictable, is the work I enjoy most.

In my PhD, I developed a hybrid-actuated shoulder exoskeleton from a whiteboard idea to tested hardware. Instead of relying on a large motor, I paired a parallel spring with a small 6 Nm motor so the spring carries the gravity load. I designed the mechanism in SolidWorks, iterated on FEA for stiffness, strength and weight, built the prototypes hands-on from machined and 3D-printed parts, and tested them. The design cut energy consumption by more than 25% and won the Best Paper Award at IFToMM ISRM 2026.

Testing and understanding failures is central to how I work. I build custom test rigs, and I validated my stiffness models of variable-stiffness actuators against hardware to within 10% RMS error before committing to fabrication. I also validated the device with users in a 12-participant study. On a 5-bar parallel robot, I questioned whether a force/torque sensor was needed at all and replaced it with force estimation from the actuator itself.

My curiosity has always gone beyond my main work. I designed and built a Go-Kart from the ground up, helped build a human-electric hybrid vehicle for the Efficycle competition, and during my Master's I made pneumatic artificial muscles from raw materials to understand how they behave under load.

So far my testing has been functional and validation testing rather than fatigue and durability testing to ISO standards, and I have not yet taken a product through industrialisation and launch. Those are exactly the steps I want to master next, and Levitate's products are where I would most like to learn them.

I would welcome the opportunity to take on your practical engineering interview and development case. References are available upon request.

Kind regards,
Arunabha Majumder